

Explicit graph invariants and exact Jacobian certificates for a self-dual five-form in ten dimensions

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Abstract

We give a reproducible certificate for 81 explicit contractions of a real self-dual five-form in signature $(1, 9)$. The fixed selection consists of one quartic, two sextic, six octic, twelve degree-ten, and sixty degree-twelve polynomials. Their Jacobian has a nonzero 81×81 minor at four exact finite-field specializations, using the same ordered functions in every case. An integral coordinate convention makes each nonzero minor a characteristic-zero independence witness when its entries are verified as derivatives of the specified contractions. An independent implementation checks all Jacobian entries at two further points, and an integral orbit-action certificate supplies the matching upper bound. This gives a maximal generically independent family. It does not determine all polynomial generators or syzygies.

1 Scope and conventions

Let F be an alternating five-form, with metric $\eta = \text{diag}(-1, 1, \dots, 1)$ and Hodge convention $\star F = F$. The self-dual subspace has dimension 126. The literature supplies the generic quotient dimension 81 and the low-degree Hilbert coefficients [1, 2]. The present certificates check the orbit dimension and matching lower bound using explicit graph contractions. The contraction catalogs and previously discovered bases are inherited from the repository's research branches; the certificate reported here re-evaluates those functions at new points. Graph enumeration and tensor-network contraction followed by numerical linear algebra are established methods for tensor invariants: see Elamaran, Ferko and Scarlett [3], whose six-dimensional three-form study leaves the ten-dimensional chiral five-form application for future work. We make no claim of priority for this method or for an 81-invariant family.

A graph has one vertex per copy of F , no loops, and valence five. An edge of multiplicity m contracts m index pairs with inverse metrics. Vertices and edges are ordered; slot order is fixed by lexicographic edge order. Graph isomorphism alone does not fix the sign of an odd-form contraction, so the formula identifier hashes the ordered edge record. Disconnected contractions are products. The quadratic contraction vanishes on this self-dual space: $F \wedge \star F = F \wedge F = 0$, since F has odd degree, so its metric norm vanishes. Complete contraction with the invariant metric proves Lorentz invariance of each graph polynomial.

For each increasing five-tuple I containing 0, define $A_I = F_I$ and let e_I denote its compact alternating component vector. The integral parametrization is

$$F(A) = \sum_{I \ni 0} A_I (e_I + \star e_I).$$

The 126 columns contain only 0, 1, -1 ; their components indexed by $I \ni 0$ form an identity matrix. Every graph polynomial therefore has integer coefficients in A , with no projector denominators.

2 Fixed selection and exact witness

The nine low-degree functions are those in `results/10d_order8.json`; the next twelve come from `results/10d_order10.json`. The first sixty selected functional directions in `results/10d_order12.json` complete the list. The inherited inventory labels two further degree-twelve directions as primitive polynomial directions; they are not included in the 81-function Jacobian. That label requires the separate homogeneous-space evidence below. A dependent Jacobian row does not identify a polynomial syzygy.

Field degree	New functional directions	Cumulative rank
4	1	1
6	2	3
8	6	9
10	12	21
12	60	81

Write $J_{aI} = \partial G_a / \partial A_I$. Each certificate cell records the prime, all 126 coordinates, all 252 self-dual compact components, the same ordered 81 graph identifiers, all entries of the 81×126 Jacobian, 81 pivot columns, and the determinant of that minor.

Prime	Seed	Rank	$\det J_* \bmod p$
32749	20260907	81	20345
32749	20260908	81	30761
32719	20260907	81	1653
32719	20260908	81	2167

The determinant is recomputed by Python-integer elimination independently of the streaming rank sieve. Euler’s identity $\sum_I A_I J_{aI} = \deg(G_a) G_a$ is checked for every row.

Suppose the recorded entries have been verified as the Jacobian of the specified integral graph polynomials at the recorded point. A nonzero minor modulo p then proves that the corresponding integer determinant polynomial is not identically zero. In characteristic zero the Jacobian criterion gives algebraic independence of these 81 polynomials, and their differentials are independent on the nonempty Zariski-open locus where that minor is nonzero. Invariance and the generic orbit dimension supply the matching upper bound. The independent audit companion `results/audit/root-review/orbit_tangent_certificate.json` records the 45 infinitesimal Lorentz generators acting on the 126 integral self-dual coordinates, using a separate signed-component implementation. Its 45×45 minors are nonzero at primes 50021 and 32749, with residues 5222 and 10852, respectively. The orbit-action entries are integral linear polynomials in A , so the same nonzero-minor lifting argument proves generic orbit dimension at least 45 in characteristic zero. There are only 45 generators, hence the generic dimension is exactly 45. Every invariant differential annihilates this tangent space, giving the upper bound $126 - 45 = 81$. The companion additionally checks that all 81 computed Jacobian rows annihilate all 45 action rows at both points. This supplies a direct witness for the generic dimension also cited in the literature. At two additional points generated after the freeze, the same ordered graph family has nonzero 81×81 minor residues 35642 modulo 50021 and 22345 modulo 32749. An independently authored forward-dual and reverse-adjoint implementation checks all 162 values and all $162 \cdot 126 = 20,412$ Jacobian entries. Integer Bareiss elimination independently verifies both minors. These implementations share NumPy arithmetic primitives but import no production contraction, Hodge, projection, rank-sieve or verifier code. On a regular local

quotient the selected invariants give local coordinates. Neither global orbit separation, generation of the invariant field by rational functions in this list, nor polynomial generation of the entire ring follows.

3 Polynomial spaces and literature comparison

The homogeneous-space dimensions reported in equation (4.2) of [1] through degree twelve are 1, 2, 7, 14, 72. These representation-theoretic counts are an external input here. Products must be included in value-space comparisons. At degree twelve the ten product monomials are I_4^3 , the three symmetric products of sextics, and the six products $I_4 I_{8,j}$. The independent audit recomputes this list and the 62 connected graph directions. Concatenating their gradients at two fresh points within each field gives a 72×252 matrix, with nonzero 72×72 minor residues 43334 modulo 50021 and 12699 modulo 32749. The integer-minor lifting argument proves linear independence of the 72 homogeneous polynomials in characteristic zero. The matching published Hilbert coefficient supplies completeness. The ten independent products therefore leave 62 primitive quotient directions. This does not assert algebraic independence of the degree-twelve list. Polynomial-space evidence uses gradients concatenated at several points; the functional Jacobian uses one point. These are separate sieves. The prose of section 4.1.4 of [1] says 64 new degree-twelve directions, inconsistent with its equation (4.2), which has exponent 62. The inventory here uses the latter; we do not treat the inconsistent prose as an additional verified count. Factor exponents in a Hilbert product are not automatically generator counts at higher degrees when generators and relations may coexist.

The octic comparison uses a selection of six displayed literature contractions: the first five expressions from equations (4.12)–(4.15) of [1], together with the hatted expression (4.18). The machine-readable matrix has exact rational entries and includes the quartic-square product explicitly. The fresh source certificate described below recovers the same matrix as the archived octic map. The 6×6 block compares primitive classes modulo products; any product correction is retained in the full 7×7 matrix.

A source audit found that the antisymmetrizers in the archived implementations of candidates J_{10} – J_{12} differ from the primary PDF and TeX source. The corrected implementation applies a red antisymmetrization of the trailing pair in the first factor of J_{10} , and of the trailing triple in each of the final three factors of J_{11} and J_{12} . These act after the black stage and before index raising. The archived readings and atlas remain separately identified; the source audit failure is preserved.

An independent source evaluator computes the corrected twelve decics and all twelve displayed octics. The exact certificate uses 14 common binary self-dual coordinate points under seven primes. Proven denominator and absolute-value bounds make signed Chinese remaindering uniquely recover the integer numerator of each rational evaluation; the combined modulus exceeds twice every bound. Exact graph evaluation matrices have ranks 7 and 14 in degrees eight and ten. Since these equal the published homogeneous space dimensions, evaluation is injective on those invariant spaces. Exact rational solving thus determines polynomial coordinates, conditional on those published upper bounds and analytic tensor invariance. A second integer/Fraction verifier checks every residue, bound, minor and map; 83 dense finite-field points provide further implementation checks. This argument uses injectivity and integer bounds, not finite holdouts alone. The complete evidence is in `results/audit/source-review/exact-source-evaluation-certificate.json`.

The corrected decic 12×14 map has rank twelve. Its union with the two product directions has rank thirteen, so the intersection dimension is one and its primitive quotient image has dimension

eleven. In the recorded graph normalization, an exact witness is

$$J_6 + \frac{5}{6}J_5 - \frac{1}{20}J_2 - \frac{11}{1008}J_1 = -\frac{13}{63}I_{4,1}I_{6,1}.$$

Consequently these twelve source expressions do not supply an invertible 12×12 map to the twelve graph primitive classes. This corrected conclusion has fresh evidence; it does not validate the archived J10–J12 transcription. The graph-only nonzero-minor proof has separate inputs.

4 Reproduction and validation

Use the declared dependencies and run the regression suite before evaluating certificates. The original low-degree commit is preserved by `v0.1-low-degree-complete`. The release commands are:

```
python scripts/graph_to_latex.py
python scripts/search_rank81.py
python scripts/verify_rank81.py
python scripts/verify_rank81.py --recompute
python scripts/verify_independent_audit.py
python scripts/validate_rank81_lorentz.py
python scripts/map_literature_basis.py --degree 8
python scripts/map_literature_basis.py --degree 10
```

The default verifier checks the saved matrix and metadata; the explicit `--recompute` mode repeats the graph evaluations with the production evaluator. The separate independent audit checker verifies saved integer minors and source CRT certificates without recomputing tensor contractions. The source-map commands verify the corrected certificate by default; historical fitting requires an explicit legacy option and a separate output. These modes are reported separately. The search stage checkpoints each raw row and its pivot, checks run identity on resume, and bounds worker submissions. Degree-specific discovery remains available through the existing nauty pipelines. `run_degree.py` accepts additional edge-list JSONL streams and separates polynomial completeness from functional rank.

Regression tests cover the 6D reproduction, self-duality, the zero quadratic contraction, exact integer arithmetic, reverse Jacobians against an independent oracle, graph parsing beyond ten vertices, and genuine Lorentz boosts. The independent audit additionally checks every selected graph under four new spatial rotations and four genuine boosts, across two primes and fresh points. These checks use exact reduced arithmetic without floating-point tolerances. They are implementation checks at specified inputs, while invariance itself follows from the complete-contraction definition.

References

- [1] M. Cederwall, J. Hutomo, S. M. Kuzenko, K. Lechner, and D. P. Sorokin, *Some remarks on invariants*, [arXiv:2509.14350v2](#).
- [2] J. Hutomo, K. Lechner, and D. P. Sorokin, *On non-linear chiral 4-form theories in D=10*, [arXiv:2509.14351](#).
- [3] A. Elamaran, C. Ferko and S. Scarlett, *Machine Learning Invariants of Tensors*, Phys. Rev. D 114 (2026) 026016, [doi:10.1103/ny3m-drnj](#), [arXiv:2512.23750](#).

A Ordered Einstein-index formulas

Each repeated index labels exactly one graph edge; its upper and lower occurrences supply one inverse metric. The machine-readable companion `results/rank81_basis.json` gives the edge lists, adjacency matrices, formula hashes, and contraction costs.

I4_1

$$F_{\mu_0,2,1\mu_0,2,2\mu_0,2,3\mu_0,3,1\mu_0,3,2} F_{\mu_1,2,1\mu_1,2,2\mu_1,3,1\mu_1,3,2\mu_1,3,3} \\ F^{\mu_0,2,1\mu_0,2,2\mu_0,2,3\mu_1,2,1\mu_1,2,2} F^{\mu_0,3,1\mu_0,3,2\mu_1,3,1\mu_1,3,2\mu_1,3,3}$$

I6_1

$$F_{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,4,1\mu_0,4,2} F_{\mu_1,3,1\mu_1,3,2\mu_1,5,1\mu_1,5,2\mu_1,5,3} \\ F_{\mu_2,4,1\mu_2,4,2\mu_2,4,3\mu_2,5,1\mu_2,5,2} F^{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_1,3,1\mu_1,3,2} \\ F^{\mu_0,4,1\mu_0,4,2\mu_2,4,1\mu_2,4,2\mu_2,4,3} F^{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_2,5,1\mu_2,5,2}$$

I6_2

$$F_{\mu_0,2,1\mu_0,2,2\mu_0,3,1\mu_0,3,2\mu_0,5,1} F_{\mu_1,3,1\mu_1,3,2\mu_1,4,1\mu_1,4,2\mu_1,5,1} \\ F_{\mu_2,4,1\mu_2,4,2\mu_2,5,1} F^{\mu_0,3,1\mu_0,3,2\mu_1,3,1\mu_1,3,2} \\ F^{\mu_1,4,1\mu_1,4,2\mu_2,4,1\mu_2,4,2} F^{\mu_0,5,1\mu_1,5,1\mu_2,5,1\mu_3,5,1\mu_4,5,1}$$

I8_1

$$F_{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,4,4\mu_0,5,1} F_{\mu_1,4,1\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_1,6,4} \\ F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,7,1} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2\mu_3,7,3\mu_3,7,4} \\ F^{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,4,4\mu_1,4,1} F^{\mu_0,5,1\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\ F^{\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_1,6,4\mu_3,6,1} F^{\mu_2,7,1\mu_3,7,1\mu_3,7,2\mu_3,7,3\mu_3,7,4}$$

I8_2

$$F_{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,3,4\mu_0,6,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,4,3\mu_1,4,4\mu_1,6,1} \\ F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,6,1} F_{\mu_3,7,1}^{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,3,4} \\ F_{\mu_4,7,1}^{\mu_1,4,1\mu_1,4,2\mu_1,4,3\mu_1,4,4} F_{\mu_5,7,1}^{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\ F_{\mu_6,7,1}^{\mu_0,6,1\mu_1,6,1\mu_2,6,1} F^{\mu_3,7,1\mu_4,7,1\mu_5,7,1\mu_6,7,1\mu_6,7,2}$$

I8_3

$$F_{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,3,4\mu_0,4,1} F_{\mu_1,4,1\mu_1,5,1\mu_1,6,1\mu_1,6,2\mu_1,7,1} \\ F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,6,1} F_{\mu_3,7,1}^{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,3,4} \\ F_{\mu_4,6,1\mu_4,7,1\mu_4,7,2}^{\mu_1,4,1\mu_1,4,1} F^{\mu_1,5,1\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\ F_{\mu_6,7,1}^{\mu_1,6,1\mu_1,6,2\mu_2,6,1\mu_4,6,1} F^{\mu_1,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2\mu_6,7,1}$$

I8_4

$$F_{\mu_0,3,1\mu_0,3,2\mu_0,6,1\mu_0,6,2\mu_0,7,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,4,3\mu_1,4,4\mu_1,6,1} \\ F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,6,1} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2}^{\mu_0,3,1\mu_0,3,2} \\ F_{\mu_4,7,1}^{\mu_1,4,1\mu_1,4,2\mu_1,4,3\mu_1,4,4} F_{\mu_5,7,1}^{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\ F^{\mu_0,6,1\mu_0,6,2\mu_1,6,1\mu_2,6,1\mu_3,6,1} F^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1\mu_5,7,1}$$

I8_5

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,3,2\mu_0,4,1\mu_0,6,1\mu_0,6,2} F_{\mu_1,4,1\mu_1,4,2\mu_1,5,1\mu_1,7,1\mu_1,7,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,6,1} F_{\mu_3,6,1\mu_3,6,2\mu_3,7,1}^{\mu_0,3,1\mu_0,3,2} \\
& F_{\mu_4,7,1\mu_4,7,2}^{\mu_0,4,1\mu_1,4,1\mu_1,4,2} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4}^{\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_3,6,1\mu_3,6,2} F_{\mu_1,7,1\mu_1,7,2\mu_3,7,1\mu_4,7,1\mu_4,7,2}^{\mu_1,7,1}
\end{aligned}$$

I8_6

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,3,2\mu_0,6,1\mu_0,6,2\mu_0,7,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,5,1\mu_1,6,1\mu_1,6,2} \\
& F_{\mu_2,4,1\mu_2,5,1\mu_2,5,2\mu_2,7,1\mu_2,7,2} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2}^{\mu_0,3,1\mu_0,3,2} \\
& F_{\mu_4,5,1\mu_4,5,2}^{\mu_1,4,1\mu_1,4,2\mu_2,4,1} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2\mu_4,5,1\mu_4,5,2}^{\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_1,6,1\mu_1,6,2\mu_3,6,1} F_{\mu_0,7,1\mu_2,7,1\mu_2,7,2\mu_3,7,1\mu_3,7,2}^{\mu_0,7,1}
\end{aligned}$$

I10_1

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,7,1\mu_0,9,1} F_{\mu_1,5,1\mu_1,6,1\mu_1,6,2\mu_1,8,1\mu_1,8,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,7,1\mu_2,7,2\mu_2,9,1} F_{\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_3,8,1\mu_3,9,1}^{\mu_2,5,1} \\
& F_{\mu_4,8,1\mu_4,8,2}^{\mu_0,4,1\mu_0,4,2\mu_0,4,3} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2}^{\mu_1,5,1} \\
& F_{\mu_1,6,1\mu_1,6,2\mu_3,6,1\mu_3,6,2\mu_3,6,3} F_{\mu_0,7,1\mu_2,7,1\mu_2,7,2\mu_5,7,1}^{\mu_0,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_0,9,1\mu_2,9,1\mu_3,9,1\mu_5,9,1\mu_7,9,1}^{\mu_0,9,1}
\end{aligned}$$

I10_2

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,6,2\mu_0,9,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,8,1\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,9,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1}^{\mu_2,6,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,7,3\mu_4,8,1\mu_4,8,2} F_{\mu_0,5,1\mu_0,5,2\mu_1,5,1\mu_1,5,2\mu_1,5,3}^{\mu_0,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_2,6,3} F_{\mu_3,7,1\mu_3,7,2\mu_4,7,1\mu_4,7,2\mu_4,7,3}^{\mu_3,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_0,9,1\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1}^{\mu_1,9,1}
\end{aligned}$$

I10_3

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,7,1\mu_0,7,2} F_{\mu_1,5,1\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_1,9,1} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,8,1\mu_2,8,2\mu_2,9,1} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1}^{\mu_2,5,1} \\
& F_{\mu_4,8,1\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2\mu_0,4,3} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2}^{\mu_1,5,1} \\
& F_{\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_3,6,1} F_{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_3,7,2\mu_5,7,1}^{\mu_0,7,1} \\
& F_{\mu_2,8,1\mu_2,8,2\mu_3,8,1\mu_4,8,1\mu_5,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_4,9,1\mu_6,9,1}^{\mu_1,9,1}
\end{aligned}$$

I10_4

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,6,1\mu_0,7,1\mu_0,7,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,7,1\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,6,4\mu_2,8,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,9,2}^{\mu_2,6,1} \\
& F_{\mu_4,8,1\mu_4,8,2\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3}^{\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,6,4} F_{\mu_0,7,1\mu_0,7,2\mu_1,7,1\mu_3,7,1\mu_3,7,2}^{\mu_0,7,1} \\
& F_{\mu_2,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_5,8,1} F_{\mu_1,9,1\mu_3,9,1\mu_3,9,2\mu_4,9,1\mu_5,9,1}^{\mu_1,9,1}
\end{aligned}$$

I10_5

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,6,1\mu_0,7,1\mu_0,9,1} F_{\mu_1,5,1\mu_1,7,1\mu_1,8,1\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,7,1\mu_2,7,2} F_{\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_3,8,1\mu_3,8,2}^{\mu_2,5,1} \\
& F_{\mu_4,6,1\mu_4,8,1\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2\mu_2,5,3}^{\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_4,6,1} F_{\mu_0,7,1\mu_1,7,1\mu_2,7,1\mu_2,7,2}^{\mu_0,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_5,8,1} F_{\mu_0,9,1\mu_1,9,1\mu_1,9,2\mu_4,9,1\mu_7,9,1}^{\mu_0,9,1}
\end{aligned}$$

I10_6

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,6,1\mu_0,8,1\mu_0,9,1} F_{\mu_1,5,1\mu_1,6,1\mu_1,8,1\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,7,1\mu_2,7,2\mu_2,8,1} F_{\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_3,7,1\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2} F_{\mu_5,7,1\mu_5,9,1}^{\mu_1,5,1\mu_2,5,1\mu_2,5,2} \\
& F_{\mu_0,6,1\mu_1,6,1\mu_3,6,1\mu_3,6,2\mu_3,6,3} F_{\mu_2,7,1\mu_2,7,2\mu_3,7,1\mu_4,7,1\mu_5,7,1} \\
& F_{\mu_0,8,1\mu_1,8,1\mu_2,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_0,9,1\mu_1,9,1\mu_1,9,2\mu_3,9,1\mu_5,9,1}
\end{aligned}$$

I10_7

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,5,1\mu_0,5,2\mu_0,7,1\mu_0,9,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,6,1\mu_1,8,1\mu_1,8,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,6,4\mu_2,7,1} F_{\mu_3,5,1\mu_3,5,2\mu_3,8,1\mu_3,9,1}^{\mu_0,3,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,9,1}^{\mu_1,4,1\mu_1,4,2} F_{\mu_5,9,1}^{\mu_0,5,1\mu_0,5,2\mu_3,5,1\mu_3,5,2} \\
& F_{\mu_1,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,6,4} F_{\mu_7,8,1\mu_7,9,1}^{\mu_0,7,1\mu_2,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1\mu_7,8,1} F_{\mu_0,9,1\mu_3,9,1\mu_4,9,1\mu_5,9,1\mu_7,9,1}
\end{aligned}$$

I10_8

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,7,1\mu_0,9,1\mu_0,9,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,5,4\mu_1,8,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,7,1\mu_2,8,1\mu_2,8,2} F_{\mu_3,6,1\mu_3,6,2\mu_3,7,1\mu_3,7,2\mu_3,7,3} \\
& F_{\mu_4,8,1\mu_4,8,2\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2} F_{\mu_5,9,1}^{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,5,4} \\
& F_{\mu_6,9,1}^{\mu_2,6,1\mu_2,6,2\mu_3,6,1\mu_3,6,2} F_{\mu_0,7,1\mu_2,7,1\mu_3,7,1\mu_3,7,2\mu_3,7,3}^{\mu_0,7,1\mu_2,7,1\mu_3,7,1\mu_3,7,2\mu_3,7,3} \\
& F_{\mu_1,8,1\mu_2,8,1\mu_2,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_0,9,1\mu_0,9,2\mu_4,9,1\mu_5,9,1\mu_6,9,1}
\end{aligned}$$

I10_9

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,7,1\mu_0,9,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,6,1\mu_1,6,2\mu_1,6,3} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,7,1\mu_2,8,1\mu_2,9,1} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2} \\
& F_{\mu_4,8,1\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2\mu_0,4,3} F_{\mu_5,9,1}^{\mu_1,5,1\mu_1,5,2\mu_2,5,1\mu_2,5,2} \\
& F_{\mu_6,9,1}^{\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_3,6,1} F_{\mu_7,8,1}^{\mu_0,7,1\mu_2,7,1\mu_3,7,1\mu_3,7,2} \\
& F_{\mu_2,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_7,8,1} F_{\mu_0,9,1\mu_2,9,1\mu_4,9,1\mu_5,9,1\mu_6,9,1}
\end{aligned}$$

I10_10

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,5,1\mu_0,5,2\mu_0,8,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,7,1\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,7,1\mu_2,8,1\mu_2,8,2\mu_2,9,1} F_{\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_3,8,1\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2} F_{\mu_5,9,1}^{\mu_0,5,1\mu_0,5,2\mu_1,5,1\mu_1,5,2\mu_1,5,3} \\
& F_{\mu_6,8,1}^{\mu_2,6,1\mu_3,6,1\mu_3,6,2\mu_3,6,3} F_{\mu_7,9,1}^{\mu_1,7,1\mu_2,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_0,8,1\mu_2,8,1\mu_2,8,2\mu_3,8,1\mu_6,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_4,9,1\mu_7,9,1}
\end{aligned}$$

I10_11

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,9,1} F_{\mu_1,4,1\mu_1,6,1\mu_1,6,2\mu_1,7,1\mu_1,7,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,8,1} F_{\mu_3,6,1\mu_3,8,1\mu_3,8,2\mu_3,9,1}^{\mu_0,3,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,9,1}^{\mu_1,4,1} F_{\mu_5,9,1}^{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_1,6,1\mu_1,6,2\mu_3,6,1} F_{\mu_0,7,1\mu_1,7,1\mu_1,7,2\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_8,9,1}^{\mu_2,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_0,9,1\mu_3,9,1\mu_4,9,1\mu_5,9,1\mu_8,9,1}
\end{aligned}$$

I10_12

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,5,1\mu_0,5,2\mu_0,8,1\mu_0,9,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,6,1\mu_1,7,1\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,7,1\mu_2,8,1\mu_2,8,2} F_{\mu_3,5,1\mu_3,5,2\mu_3,8,1\mu_3,9,1}^{\mu_0,3,1} \\
& F_{\mu_4,6,1\mu_4,7,1\mu_4,7,2}^{\mu_1,4,1\mu_1,4,2} F_{\mu_5,8,1}^{\mu_0,5,1\mu_0,5,2\mu_3,5,1\mu_3,5,2} \\
& F_{\mu_6,9,1}^{\mu_1,6,1\mu_2,6,1\mu_2,6,2\mu_4,6,1} F_{\mu_7,9,1}^{\mu_1,7,1\mu_2,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_0,8,1\mu_2,8,1\mu_2,8,2\mu_3,8,1\mu_5,8,1} F^{\mu_0,9,1\mu_1,9,1\mu_3,9,1\mu_6,9,1\mu_7,9,1}
\end{aligned}$$

I12_1

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F^{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F^{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_2

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F^{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_3

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_4

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F^{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_8,11,1\mu_8,11,2}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1} F^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_8,11,1\mu_8,11,2}
\end{aligned}$$

I12_5

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_6

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_7

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_8

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,7,3\mu_4,8,1\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2\mu_4,7,3} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_2,11,1\mu_3,11,1\mu_8,11,1\mu_8,11,2}^{\mu_1,11,1\mu_2,11,1\mu_3,11,1\mu_8,11,1\mu_8,11,2}
\end{aligned}$$

I12_9

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_2,9,2\mu_3,9,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_10

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1}^{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_11

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,9,3} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_2,11,1}^{\mu_2,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_12

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1}^{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_13

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,10,2\mu_4,11,1} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1}^{\mu_0,5,1\mu_0,5,2\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1}^{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2\mu_9,11,1}
\end{aligned}$$

I12_14

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,9,3} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_2,11,1}^{\mu_2,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2}
\end{aligned}$$

I12_15

$$\begin{aligned}
& F_{\mu_{0,5,1}\mu_{0,6,1}\mu_{0,7,1}\mu_{0,7,2}\mu_{0,10,1}} F_{\mu_{1,5,1}\mu_{1,8,1}\mu_{1,9,1}\mu_{1,9,2}\mu_{1,11,1}} \\
& F_{\mu_{2,6,1}\mu_{2,6,2}\mu_{2,6,3}\mu_{2,9,1}\mu_{2,10,1}} F_{\mu_{3,7,1}\mu_{3,8,1}\mu_{3,9,1}\mu_{3,11,1}\mu_{3,11,2}} \\
& F_{\mu_{4,7,1}\mu_{4,8,1}\mu_{4,8,2}\mu_{4,8,3}\mu_{4,10,1}} F_{\mu_{5,6,1}\mu_{5,10,1}\mu_{5,10,2}}^{\mu_{0,5,1}\mu_{1,5,1}} \\
& F_{\mu_{0,6,1}\mu_{2,6,1}\mu_{2,6,2}\mu_{2,6,3}\mu_{5,6,1}}^{\mu_{0,7,1}\mu_{0,7,2}\mu_{3,7,1}\mu_{4,7,1}} F_{\mu_{7,11,1}}^{\mu_{0,7,1}\mu_{0,7,2}\mu_{3,7,1}\mu_{4,7,1}} \\
& F_{\mu_{1,8,1}\mu_{3,8,1}\mu_{4,8,1}\mu_{4,8,2}\mu_{4,8,3}} F_{\mu_{9,11,1}}^{\mu_{1,9,1}\mu_{1,9,2}\mu_{2,9,1}\mu_{3,9,1}} \\
& F_{\mu_{0,10,1}\mu_{2,10,1}\mu_{4,10,1}\mu_{5,10,1}\mu_{5,10,2}} F_{\mu_{1,11,1}\mu_{3,11,1}\mu_{3,11,2}\mu_{7,11,1}\mu_{9,11,1}}
\end{aligned}$$

I12_16

$$\begin{aligned}
& F_{\mu_{0,5,1}\mu_{0,6,1}\mu_{0,6,2}\mu_{0,7,1}\mu_{0,10,1}} F_{\mu_{1,5,1}\mu_{1,8,1}\mu_{1,9,1}\mu_{1,11,1}\mu_{1,11,2}} \\
& F_{\mu_{2,6,1}\mu_{2,9,1}\mu_{2,9,2}\mu_{2,10,1}\mu_{2,10,2}} F_{\mu_{3,7,1}\mu_{3,8,1}\mu_{3,8,2}\mu_{3,9,1}\mu_{3,11,1}} \\
& F_{\mu_{4,7,1}\mu_{4,7,2}\mu_{4,8,1}\mu_{4,8,2}\mu_{4,10,1}} F_{\mu_{5,6,1}\mu_{5,6,2}\mu_{5,10,1}}^{\mu_{0,5,1}\mu_{1,5,1}} \\
& F_{\mu_{0,6,1}\mu_{0,6,2}\mu_{2,6,1}\mu_{5,6,1}\mu_{5,6,2}}^{\mu_{0,7,1}\mu_{3,7,1}\mu_{4,7,1}\mu_{4,7,2}} F_{\mu_{7,11,1}}^{\mu_{0,7,1}\mu_{3,7,1}\mu_{4,7,1}\mu_{4,7,2}} \\
& F_{\mu_{1,8,1}\mu_{3,8,1}\mu_{3,8,2}\mu_{4,8,1}\mu_{4,8,2}} F_{\mu_{9,11,1}}^{\mu_{1,9,1}\mu_{2,9,1}\mu_{2,9,2}\mu_{3,9,1}} \\
& F_{\mu_{0,10,1}\mu_{2,10,1}\mu_{2,10,2}\mu_{4,10,1}\mu_{5,10,1}} F_{\mu_{1,11,1}\mu_{1,11,2}\mu_{3,11,1}\mu_{7,11,1}\mu_{9,11,1}}
\end{aligned}$$

I12_17

$$\begin{aligned}
& F_{\mu_{0,5,1}\mu_{0,6,1}\mu_{0,7,1}\mu_{0,10,1}\mu_{0,10,2}} F_{\mu_{1,5,1}\mu_{1,5,2}\mu_{1,8,1}\mu_{1,9,1}\mu_{1,11,1}} \\
& F_{\mu_{2,6,1}\mu_{2,6,2}\mu_{2,6,3}\mu_{2,9,1}\mu_{2,10,1}} F_{\mu_{3,7,1}\mu_{3,8,1}\mu_{3,9,1}\mu_{3,9,2}\mu_{3,9,3}} \\
& F_{\mu_{4,7,1}\mu_{4,8,1}\mu_{4,8,2}\mu_{4,10,1}\mu_{4,11,1}} F_{\mu_{5,6,1}\mu_{5,10,1}}^{\mu_{0,5,1}\mu_{1,5,1}\mu_{1,5,2}} \\
& F_{\mu_{0,6,1}\mu_{2,6,1}\mu_{2,6,2}\mu_{2,6,3}\mu_{5,6,1}}^{\mu_{0,7,1}\mu_{3,7,1}\mu_{4,7,1}} F_{\mu_{7,11,1}\mu_{7,11,2}}^{\mu_{0,7,1}\mu_{3,7,1}\mu_{4,7,1}\mu_{4,7,2}} \\
& F_{\mu_{8,11,1}}^{\mu_{1,8,1}\mu_{3,8,1}\mu_{4,8,1}\mu_{4,8,2}} F_{\mu_{8,11,1}}^{\mu_{1,9,1}\mu_{2,9,1}\mu_{3,9,1}\mu_{3,9,2}\mu_{3,9,3}} \\
& F_{\mu_{0,10,1}\mu_{0,10,2}\mu_{2,10,1}\mu_{4,10,1}\mu_{5,10,1}} F_{\mu_{1,11,1}\mu_{4,11,1}\mu_{7,11,1}\mu_{7,11,2}\mu_{8,11,1}}
\end{aligned}$$

I12_18

$$\begin{aligned}
& F_{\mu_{0,5,1}\mu_{0,5,2}\mu_{0,6,1}\mu_{0,7,1}\mu_{0,10,1}} F_{\mu_{1,5,1}\mu_{1,8,1}\mu_{1,8,2}\mu_{1,9,1}\mu_{1,11,1}} \\
& F_{\mu_{2,6,1}\mu_{2,6,2}\mu_{2,6,3}\mu_{2,9,1}\mu_{2,10,1}} F_{\mu_{3,7,1}\mu_{3,8,1}\mu_{3,8,2}\mu_{3,9,1}\mu_{3,11,1}} \\
& F_{\mu_{4,7,1}\mu_{4,7,2}\mu_{4,8,1}\mu_{4,10,1}\mu_{4,10,2}} F_{\mu_{5,6,1}\mu_{5,10,1}}^{\mu_{0,5,1}\mu_{0,5,2}\mu_{1,5,1}} \\
& F_{\mu_{0,6,1}\mu_{2,6,1}\mu_{2,6,2}\mu_{2,6,3}\mu_{5,6,1}}^{\mu_{0,7,1}\mu_{3,7,1}\mu_{4,7,1}\mu_{4,7,2}} F_{\mu_{7,11,1}}^{\mu_{0,7,1}\mu_{3,7,1}\mu_{4,7,1}\mu_{4,7,2}} \\
& F_{\mu_{1,8,1}\mu_{1,8,2}\mu_{3,8,1}\mu_{3,8,2}\mu_{4,8,1}} F_{\mu_{9,11,1}\mu_{9,11,2}}^{\mu_{1,9,1}\mu_{2,9,1}\mu_{3,9,1}} \\
& F_{\mu_{0,10,1}\mu_{2,10,1}\mu_{4,10,1}\mu_{4,10,2}\mu_{5,10,1}} F_{\mu_{1,11,1}\mu_{3,11,1}\mu_{7,11,1}\mu_{9,11,1}\mu_{9,11,2}}
\end{aligned}$$

I12_19

$$\begin{aligned}
& F_{\mu_{0,5,1}\mu_{0,6,1}\mu_{0,6,2}\mu_{0,7,1}\mu_{0,10,1}} F_{\mu_{1,5,1}\mu_{1,8,1}\mu_{1,9,1}\mu_{1,11,1}\mu_{1,11,2}} \\
& F_{\mu_{2,6,1}\mu_{2,9,1}\mu_{2,9,2}\mu_{2,10,1}\mu_{2,10,2}} F_{\mu_{3,7,1}\mu_{3,7,2}\mu_{3,8,1}\mu_{3,9,1}\mu_{3,9,2}} \\
& F_{\mu_{4,7,1}\mu_{4,8,1}\mu_{4,8,2}\mu_{4,10,1}\mu_{4,11,1}} F_{\mu_{5,6,1}\mu_{5,6,2}\mu_{5,10,1}}^{\mu_{0,5,1}\mu_{1,5,1}} \\
& F_{\mu_{0,6,1}\mu_{0,6,2}\mu_{2,6,1}\mu_{5,6,1}\mu_{5,6,2}}^{\mu_{0,7,1}\mu_{3,7,1}\mu_{3,7,2}\mu_{4,7,1}} F_{\mu_{7,11,1}}^{\mu_{0,7,1}\mu_{3,7,1}\mu_{3,7,2}\mu_{4,7,1}} \\
& F_{\mu_{8,11,1}}^{\mu_{1,8,1}\mu_{3,8,1}\mu_{4,8,1}\mu_{4,8,2}} F_{\mu_{8,11,1}}^{\mu_{1,9,1}\mu_{2,9,1}\mu_{2,9,2}\mu_{3,9,1}\mu_{3,9,2}} \\
& F_{\mu_{0,10,1}\mu_{2,10,1}\mu_{2,10,2}\mu_{4,10,1}\mu_{5,10,1}} F_{\mu_{1,11,1}\mu_{1,11,2}\mu_{4,11,1}\mu_{7,11,1}\mu_{8,11,1}}
\end{aligned}$$

I12_20

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1\mu_9,11,2} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_7,11,2\mu_9,11,1}
\end{aligned}$$

I12_21

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,10,2} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1\mu_9,11,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_22

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_8,11,1\mu_8,11,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_23

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,10,2\mu_4,11,1} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_8,11,1\mu_8,11,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_7,11,2\mu_8,11,1}
\end{aligned}$$

I12_24

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_8,11,1\mu_8,11,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_25

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_26

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_27

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_28

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_29

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_30

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,10,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_2,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_31

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_32

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_33

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_34

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1\mu_7,11,2}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_2,11,1\mu_3,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2}
\end{aligned}$$

I12_35

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,9,3} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F^{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_36

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F^{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_37

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_38

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,9,3} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_39

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_9,11,1\mu_9,11,2} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_40

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_41

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_42

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1\mu_9,11,2}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_43

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_2,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_44

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_45

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_9,11,1}^{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_46

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,11,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,10,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_2,11,1\mu_7,11,1\mu_9,11,1}^{\mu_1,11,1\mu_1,11,2\mu_2,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_47

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_6,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_3,11,1\mu_3,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}^{\mu_3,11,1\mu_3,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_48

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,10,2} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_0,5,2\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1} F_{\mu_8,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_8,11,1\mu_8,11,2}^{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_8,11,1\mu_8,11,2}
\end{aligned}$$

I12_49

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_8,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_8,11,1}^{\mu_1,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_50

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{F_{\mu_0,5,1\mu_1,5,1}} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_3,11,1\mu_3,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_51

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{F_{\mu_0,5,1\mu_1,5,1}} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_52

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{F_{\mu_0,5,1\mu_1,5,1}} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_8,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_53

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,7,3\mu_4,8,1\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{F_{\mu_0,5,1\mu_1,5,1}} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2\mu_4,7,3} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_8,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_6,11,1\mu_8,11,1}
\end{aligned}$$

I12_54

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,10,2\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{F_{\mu_0,5,1\mu_0,5,2\mu_1,5,1}} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2\mu_9,11,1}
\end{aligned}$$

I12_55

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_56

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,7,3\mu_4,8,1\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2\mu_4,7,3} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_6,11,1\mu_8,11,1}
\end{aligned}$$

I12_57

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2\mu_8,11,1}
\end{aligned}$$

I12_58

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_59

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,6,1\mu_0,6,2\mu_0,8,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,7,1\mu_1,8,1\mu_1,9,1} \\
& F_{\mu_2,7,1\mu_2,7,2\mu_2,9,1\mu_2,10,1\mu_2,11,1} F_{\mu_3,9,1\mu_3,9,2\mu_3,10,1\mu_3,10,2\mu_3,11,1} \\
& F_{\mu_4,6,1\mu_4,9,1\mu_4,10,1} F_{\mu_5,7,1\mu_5,8,1\mu_5,11,1}^{\mu_0,4,1\mu_0,4,2} \\
& F_{\mu_6,10,1\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_4,6,1} F_{\mu_7,8,1}^{\mu_1,7,1\mu_2,7,1\mu_2,7,2\mu_5,7,1} \\
& F_{\mu_8,11,1}^{\mu_0,8,1\mu_1,8,1\mu_5,8,1\mu_7,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_4,9,1} \\
& F_{\mu_2,10,1\mu_3,10,1\mu_3,10,2\mu_4,10,1\mu_6,10,1} F_{\mu_2,11,1\mu_3,11,1\mu_5,11,1\mu_6,11,1\mu_8,11,1}
\end{aligned}$$

I12_60

$$\begin{aligned}
& F_{\mu_{0,4,1}\mu_{0,6,1}\mu_{0,6,2}\mu_{0,10,1}\mu_{0,11,1}} F_{\mu_{1,5,1}\mu_{1,7,1}\mu_{1,7,2}\mu_{1,8,1}\mu_{1,9,1}} \\
& F_{\mu_{2,7,1}\mu_{2,9,1}\mu_{2,9,2}\mu_{2,10,1}\mu_{2,11,1}} F_{\mu_{3,8,1}\mu_{3,9,1}\mu_{3,10,1}\mu_{3,10,2}\mu_{3,11,1}} \\
& F_{\mu_{0,4,1}}^{\mu_{4,6,1}\mu_{4,6,2}\mu_{4,10,1}\mu_{4,11,1}} F_{\mu_{5,7,1}\mu_{5,8,1}\mu_{5,8,2}\mu_{5,9,1}}^{\mu_{1,5,1}} \\
& F_{\mu_{0,6,1}\mu_{0,6,2}\mu_{4,6,1}\mu_{4,6,2}}^{\mu_{6,11,1}} F_{\mu_{1,7,1}\mu_{1,7,2}\mu_{2,7,1}\mu_{5,7,1}}^{\mu_{7,8,1}} \\
& F_{\mu_{1,8,1}\mu_{3,8,1}\mu_{5,8,1}\mu_{5,8,2}\mu_{7,8,1}}^{\mu_{1,9,1}\mu_{2,9,1}\mu_{2,9,2}\mu_{3,9,1}\mu_{5,9,1}} \\
& F_{\mu_{0,10,1}\mu_{2,10,1}\mu_{3,10,1}\mu_{3,10,2}\mu_{4,10,1}}^{\mu_{0,11,1}\mu_{2,11,1}\mu_{3,11,1}\mu_{4,11,1}\mu_{6,11,1}}
\end{aligned}$$